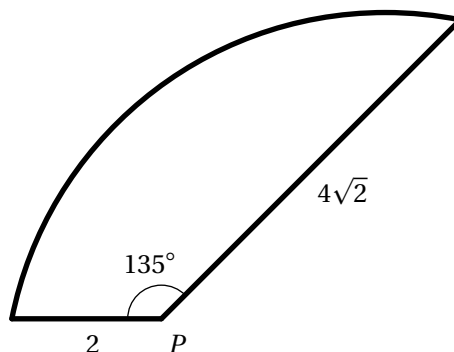


1. Rectangle $ABCD$ is drawn such that $AB = 7$ and $BC = 4$. Suppose $BDEF$ is a square that contains vertex C in its interior. Find $CE^2 + CF^2$.
2. Pictured below is a cut-out portion of a circle. Suppose that the circular arc length is one quarter of the circumference of the original circle. If the distance between P and the center of the circle is d , find d^2 .



3. Let $\triangle ABC$ be an isosceles triangle satisfying $AB = AC$ and $\angle BAC = 20^\circ$. Let P be a point on AC such that $AP = BC$. Find $\angle ADB$.
4. In $\triangle ABC$, suppose M is the midpoint of side BC , AN bisects $\angle BAC$, and $BN \perp AN$. If sides AB and AC have lengths 14 and 19, respectively, then find MN .
5. In quadrilateral $ABCD$, $\angle DAC = 98^\circ$, $\angle DBC = 82^\circ$, $\angle BCD = 70^\circ$, and $BC = AD$. Find $\angle ACD$.
6. Point P lies in the interior of rectangle $ABCD$ such that $AP + CP = 27$, $BP - DP = 17$, and $\angle DAP \cong \angle DCP$. Compute the area of rectangle $ABCD$.
7. Let $\triangle ABC$ be a triangle satisfying $\angle ABC = \angle ACB = 42^\circ$. Suppose X lies in $\triangle ABC$ such that $\angle XBC = 30^\circ$ and $\angle XCB = 18^\circ$. Compute $\angle AXB$.
8. Let $ABCD$ be a quadrilateral such that $AD = BC$. Points X , Y , and Z are the midpoints of AB , BD , and DC , respectively. In degrees, find $\angle YZX$ such that $\angle B = 81^\circ$, $\angle C = 125^\circ$, and $\angle D = 100^\circ$.